CNC Bending services

10 Materials
.030"- .250" Thick

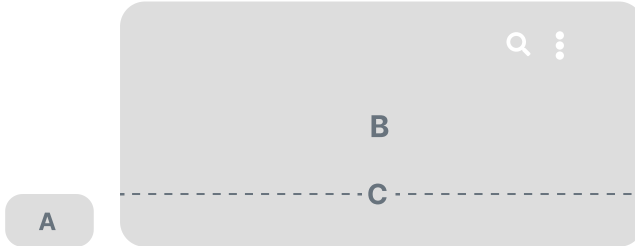
+1-2 days
Production Time

GET INSTANT PRICING (HTTPS://APP.SENDCUTSEND.COM/CUSTOMER)

BENEFITS MATERIALS GUIDELINES WHAT TO EXPECT

Bending sizing and options

PART SIZES



A. **.375" x 1.5"** Min part size

B. **30" x 44"** Max part size

C. **16" x 44"** Maximum Bend Length

10 MATERIALS, .030" - .250"

5052 H32 Aluminum | Brass | 4130 Chromoly | Copper | G90 Galvanized Steel | Mild Steel | Polycarbonate | 304 Stainless Steel | 316 Stainless Steel | Grade2 Titanium

TOLERANCE

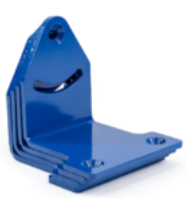
+/- 1 degree or better

Fast turnarounds to keep your projects moving

Accurate, repeatable bends

Fast Turnarounds

Simple, all-in-one workflow



*"SCS beats my local shops in every dimension**... Did some bends my usual shop would have baulked at**."*

(https://sendcutsend.com)

Available materials for Bending

One step to a finished part
Built to take abuse
Looks as good as it performs

5052 H32 Aluminum
8 thicknesses: .040" – .250"
→ (https://sendcutsend.com/materials/5052-aluminum/)

Brass
5 thicknesses: .040" – .250"
→ (https://sendcutsend.com/materials/brass/)

4130 Chromoly
5 thicknesses: .050" – 250"
→ (https://sendcutsend.com/materials/4130-chromoly/)

Copper
5 thicknesses: .040" – .250"
→ (https://sendcutsend.com/materials/copper/)

G90 Galvanized Steel
4 thicknesses: .030" – .074"
→ (https://sendcutsend.com/materials/g90-steel/)

Mild Steel
9 thicknesses: .030" – .250"
→ (https://sendcutsend.com/materials/mild-steel/)

Polycarbonate
3 thicknesses: .118" – .220"
→ (https://sendcutsend.com/materials/polycarbonate/)

304 Stainless Steel
8 thicknesses: .030" – .250"
→ (https://sendcutsend.com/materials/stainless-steel/)

316 Stainless Steel
4 thicknesses: .060" – .250"
→ (https://sendcutsend.com/materials/stainless-steel-316/)

Grade 2 Titanium
1 thickness: .040"
→ (https://sendcutsend.com/materials/titanium-cp2/)

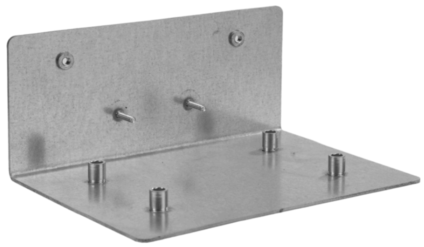
Upload now for instant pricing

Bending adds 1-2 days to production. For the fastest lead time, order bent parts separately

(https://sendcutsend.com)

GET STARTED
(HTTPS://APP.SENDCUTSEND.COM/CUSTOMER)

Affordable pricing. Up to 80% off in bulk.



G90 Galvanized Steel .059"

5.4" x 6"

Laser Cutting, Bending, Hardware Insertion

QTY 1:
$31.76/ea

QTY 100:
$8.32/ea

Cold Rolled Steel .119"

6.3" x 7.4"

Laser Cutting, Bending, Dimple Forming, Hardware Insertion, Tapping, Powder Coating

QTY 1:
$73.33/ea

QTY 100:
$16.21/ea

Polycarbonate .118"

4" x 5.5"

CNC Routing, Bending

(https://sendcutsend.com) QTY 1: $24.98/ea

QTY 100: $7.84/ea

Design guidelines for bent parts

Part size and bend limits

Sizing limits apply to flat parts **before bending (https://sendcutsend.com/services/cnc-bending/)**

Min part size (flat)	.375″ x 1.5″
Max part size (flat)	30″ x 44″
Maximum bend length	16-44″ depending on thickness
Max Bend Angle	130°

See **Processing Min/Max Size Charts (https://sendcutsend.com/materials/processing-min-max/)** for material specific limits

Use our bending specifications

Bend radius and K factor are set per material thickness. Design your parts using our specs so they form as expected, we don't offer custom bend radii.

Find specs in our **Bending Calculator (https://sendcutsend.com/bending-calculator/)** or **Material Catalog (https://sendcutsend.com/materials/)**.

Working in CAD? Import our gauge tables for **Autodesk Fusion (https://sendcutsend.com/faq/does-sendcutsend-have-gauge-tables-for-fusion-360/)** or **SolidWorks (https://sendcutsend.com/blog/how-to-import-and-use-sheet-metal-gauge-tables-in-solidworks/)** to apply the latest specifications automatically.

File setup

Upload a 2D vector file (.dxf, .dwg, .ai, .eps) or a 3D file (.step, .stp). You'll confirm bend angles and flange orientations in a 3D preview during checkout.

SOFTWARE	FORMAT	BEND LINE
Fusion (https://sendcutsend.com/blog/how-to-export-a-dxf-in-fusion-360/)	.dxf, .step, .stp	Solid line (default)
Adobe Illustrator (https://sendcutsend.com/blog/set-up-adobe-illustrator-file-for-metal-bending/)	.ai	Solid, separate color from cut lines

(https://sendcutsend.com)

SolidWorks (https://sendcutsend.com/blog/how-to-map-bend-lines-into-layers-and-linetypes-in-solidworks/)	.dxf, .step, .stp	Dashed line (**not** hidden)
AutoCAD (https://sendcutsend.com/blog/how-to-export-from-autocad/)	.dxf, .step, .stp	Dashed line (**not** hidden)
CorelDraw	.eps	Solid, separate color from cut lines
Inkscape (https://sendcutsend.com/blog/designing-for-laser-cutting-in-inkscape/)	.eps	Solid, separate color from cut lines

2D files: Mark bend locations with a line at the center of each bend. Line color doesn't matter, our system ignores it.

3D files: No bend lines needed. Model your part with our bend specs, save, and upload. **See how to configure STEP file bends (https://sendcutsend.com/faq/how-to-configure-bends-for-my-step-file/)**.

What we can't bend

- No acute angles greater than 130°
- No obtuse angles less than 5°
- No curl, bump, or roll forming
- No coining or hemming
- No joggle bends in polycarbonate

What to expect with bent parts

All bent parts:

- Bend-to-edge tolerance: **+/- 0.015″** (each additional bend adds at least 0.015″)
- Some bulging at the ends of bends is expected
- Witness marks from the bending process will be visible, these can become deeper and more noticeable depending on the material
- We don't offer special protection for cosmetic parts at this time

Sheet metal bend angle tolerance:

- Bends up to 24″: **+/- 1°**
- Bends longer than 24″: **+/- 2°**

Polycarbonate bend angle tolerance:

- Bends up to 24″: **+/- 5°**
- Bends longer than 24″: **+/- 7°**

Polycarbonate-specific considerations:

- Adjacent flanges bent in the same direction need clearance.
- Polycarbonate is over-bent to achieve the desired angle, so we recommend **1:1 clearance** to material thickness
- Minor surface edge cracks may appear depending on material thickness and bend angle
- Noticeable cracks will appear in stressed areas if adequate bend relief is not provided

Flange requirements

Minimum flange length depends on material and thickness — check the specs on your chosen material in the **Material Catalog (https://sendcutsend.com/materials/)**.

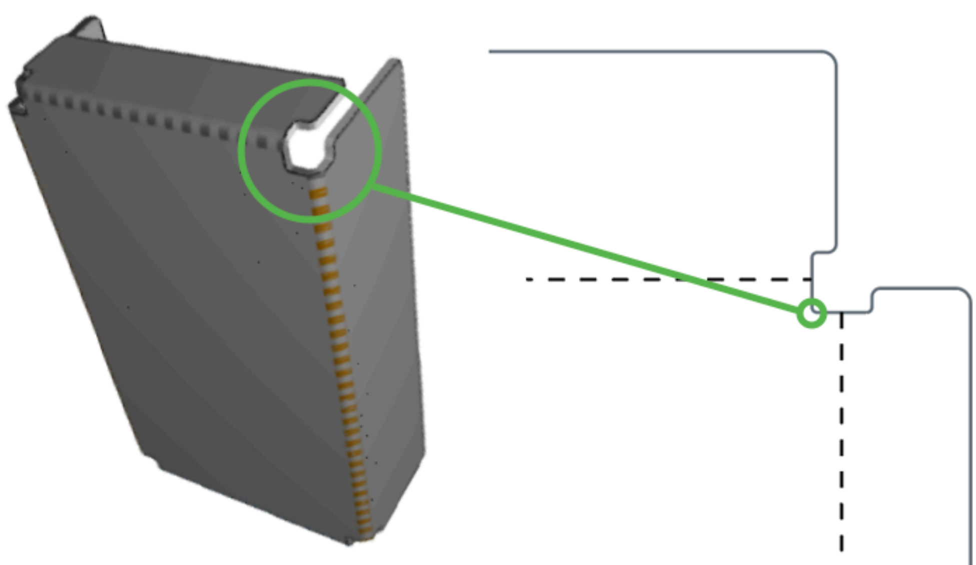
For U-channel and C-channel bends:

- Sheet metal: base must be at least **2x the flange length (2:1 ratio)**
- Polycarbonate: base must be at least **3x the flange length (3:1 ratio)**

Learn more about **channel bend exceptions for thin sheet metal (https://sendcutsend.com/faq/what-are-your-channel-bend-requirements/)**

Bend relief

Bend relief notches reduce stress on inner radii and prevent corner tearing or bulging. This is critical for polycarbonate, which cracks without adequate relief.



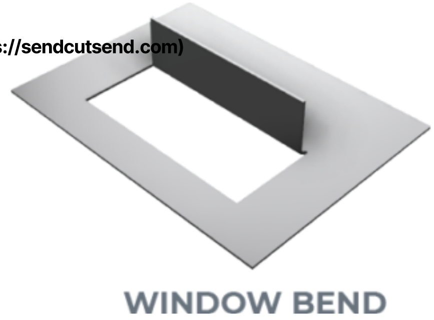
Learn more about bend relief requirements (https://sendcutsend.com/faq/what-are-your-bend-relief-requirements/)

Window and joggle bends

Window bends are allowed up to 90° for sheet metal and polycarbonate. More acute angles need review by our team.

Joggle bends are allowed up to 90° for sheet metal only. Min and max joggle flange values are on each material's spec chart.

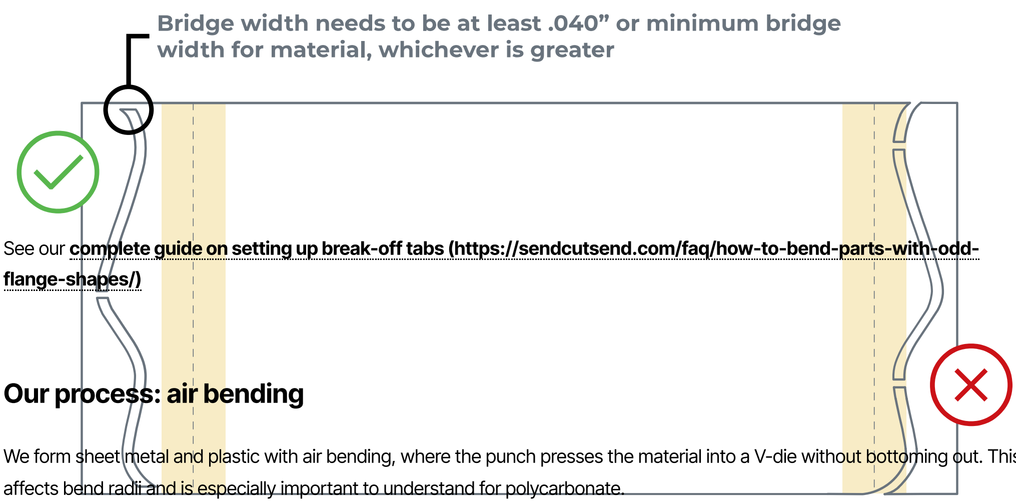
(https://sendcutsend.com)



Window bend requirements (https://sendcutsend.com/faq/what-are-your-window-bend-requirements/)
Joggle bend requirements (https://sendcutsend.com/faq/what-are-your-joggle-bend-requirements/)

Odd flange shapes

We can bend irregular flange shapes, but we need a flat surface parallel to the bend. Add break-off tabs to create a straight reference edge, then remove them after you receive your part.



See our **complete guide on setting up break-off tabs (https://sendcutsend.com/faq/how-to-bend-parts-with-odd-flange-shapes/)**

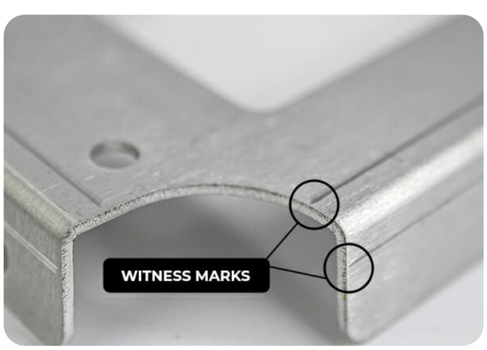
Our process: air bending

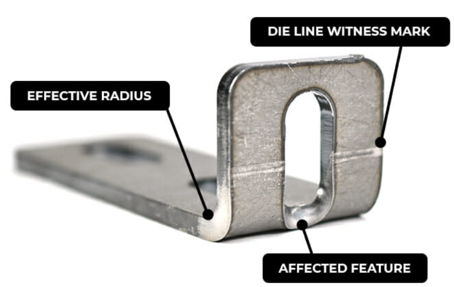
We form sheet metal and plastic with air bending, where the punch presses the material into a V-die without bottoming out. This affects bend radii and is especially important to understand for polycarbonate.

Learn more about air bending (https://sendcutsend.com/faq/how-do-you-bend-parts/)

Die lines and feature distortion

Die lines are witness marks left where the tooling makes contact with your part during forming. Cut features that fall within die lines or the bend area will distort as the material stretches.





Learn how to avoid feature distortion in our **Bending Deformation Guidelines (https://sendcutsend.com/guidelines/bend-deformation/)**

Common issues to avoid

(https://sendcutsend.com)

Combined lines: Bends on a common axis must be joined. If they're separate, each will be treated as an individual bend.

Intersecting bends: We can't bend intersecting lines that don't have separate flanges.

Incomplete bends: A bend must extend all the way across the area being bent. We can't form partial bends.

Insufficient contact: Flanges need to meet the minimum flange length for at least ~50–60% of the bend's length on both sides to provide enough press brake support. Thicker materials (.187″–.250″) need more contact.

Insufficient or missing bend relief: Certain designs require bend relief to avoid damage to the part. Without proper relief, a part cannot be bent accurately. This is critical for polycarbonate parts since the material is prone to cracking.

Additional bending resources

If you're designing parts for powder coating, these resources may help:

- **Processing min/max part sizes (https://sendcutsend.com/materials/processing-min-max/)**
- **Material min/max part sizes (https://sendcutsend.com/materials/processing-min-max/)**
- **Bending Deformation guidelines (https://sendcutsend.com/guidelines/bend-deformation/)**
- **Bending Calculator (https://sendcutsend.com/bending-calculator/)**

Pre-flight checklist

- ✓ File is in a format that we accept (2D: .dxf, .dwg, .ai, .eps; 3D: .step or .stp)
- ✓ All holes and cutouts are at least 50% material thickness for laser cut parts
- ✓ All holes and cutouts are no less than 0.070″ for most waterjet cut parts
- ✓ All holes and cutouts are no less than 0.125″ for all CNC routed parts
- ✓ File is built at a 1:1 scale, preferably in inch or mm units
- ✓ 2D files: all objects on the same layer, text converted to outlines, no open contours, all shapes united or merged, floating interiors bridged or stencilized
- ✓ 3D files: single solid sheet metal body, uniform thickness with a face at every edge, modeled at stock thickness with matching bend specs
- ✓ All stray points, duplicate lines, empty objects and text areas have been removed

SendCutSend reviews

Customers love getting their sheet metal parts from us

(HTTPS://WWW.GOOGLE.COM/SEARCH?SCA_ESV=5CF68B8F0ED15ED3&SXSRF=ANBL-N4DD3YZIZINWBQ2BOVYM7FO5MEOYQ:17700
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Top Quality Store

Bending FAQs

How do I keep perpendicular corner flanges from colliding in my design? (https://sendcutsend.com/faq/how-to-prevent-corner-flanges-from-colliding/)

READ MORE » (HTTPS://SENDCUTSEND.COM/FAQ/HOW-TO-PREVENT-CORNER-FLANGES-FROM-COLLIDING/)

What bending specifications does SendCutSend use for each material? (https://sendcutsend.com/faq/what-are-your-material-bending-specifications/)

READ MORE » (HTTPS://SENDCUTSEND.COM/FAQ/WHAT-ARE-YOUR-MATERIAL-BENDING-SPECIFICATIONS/)

Does SendCutSend have gauge tables for Autodesk Fusion? (https://sendcutsend.com/faq/does-sendcutsend-have-gauge-tables-for-fusion-360/)

READ MORE » (HTTPS://SENDCUTSEND.COM/FAQ/DOES-SENDCUTSEND-HAVE-GAUGE-TABLES-FOR-FUSION-360/)

Why doesn’t bend radius always match material thickness? (https://sendcutsend.com/faq/why-doesnt-the-bend-radius-correlate-with-material-thickness/)

READ MORE » (HTTPS://SENDCUTSEND.COM/FAQ/WHY-DOESNT-THE-BEND-RADIUS-CORRELATE-WITH-MATERIAL-THICKNESS/)

What are SendCutSend’s bending requirements? (https://sendcutsend.com/faq/what-are-your-bending-requirements/)

READ MORE » (HTTPS://SENDCUTSEND.COM/FAQ/WHAT-ARE-YOUR-BENDING-REQUIREMENTS/)

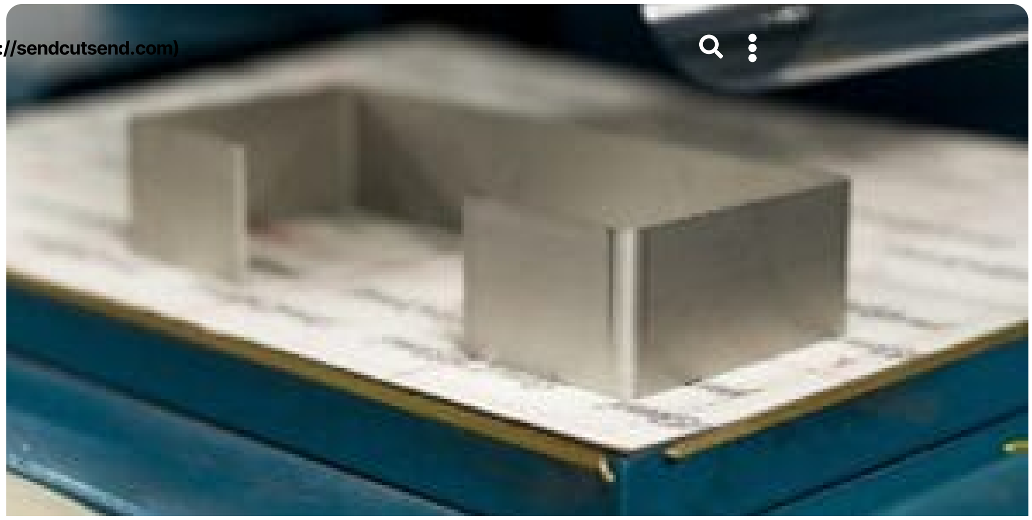
MORE FAQS → (HTTPS://SENDCUTSEND.COM/FAQ-CATEGORY/POWDER-COATING/)

Bending posts

(https://sendcutsend.com/blog/dfm-for-bending-top-sheet-metal-bending-mistakes-to-avoid/)

DFM for Bending: Top Sheet Metal Bending Mistakes to Avoid (https://sendcutsend.com/blog/dfm-for-bending-top-sheet-metal-bending-mistakes-to-avoid/)

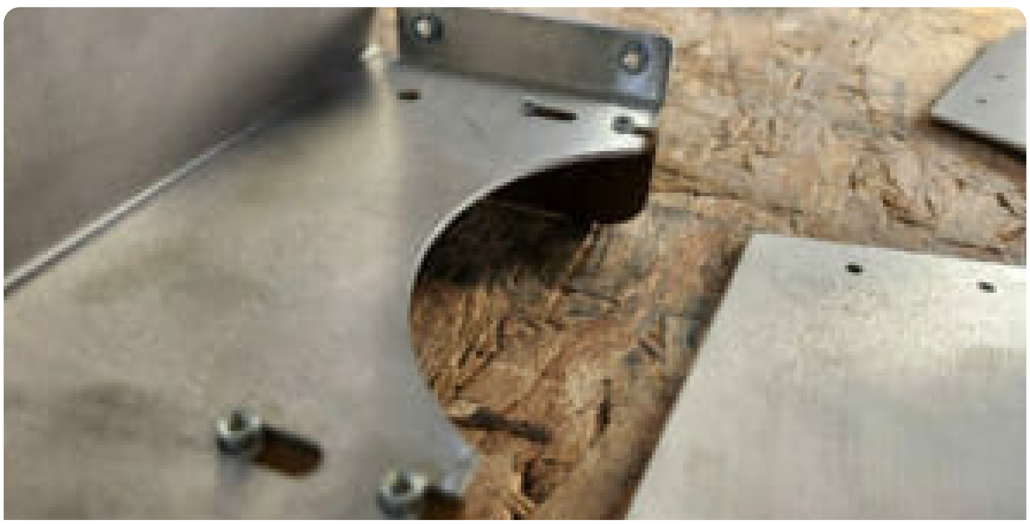
READ MORE » (HTTPS://SENDCUTSEND.COM/BLOG/DFM-FOR-BENDING-TOP-SHEET-METAL-BENDING-MISTAKES-TO-AVOID/)



(https://sendcutsend.com/blog/the-ultimate-guide-to-sheet-metal-plastic-bending/)

The Ultimate Guide to Sheet Metal & Plastic Bending: Tolerances, Requirements, and Best Practices (https://sendcutsend.com/blog/the-ultimate-guide-to-sheet-metal-plastic-bending/)

READ MORE » (HTTPS://SENDCUTSEND.COM/BLOG/THE-ULTIMATE-GUIDE-TO-SHEET-METAL-PLASTIC-BENDING/)



(https://sendcutsend.com/blog/designing-with-and-ordering-sheet-metal-hardware/)

Designing With and Ordering Sheet Metal Hardware (https://sendcutsend.com/blog/designing-with-and-ordering-sheet-metal-hardware/)

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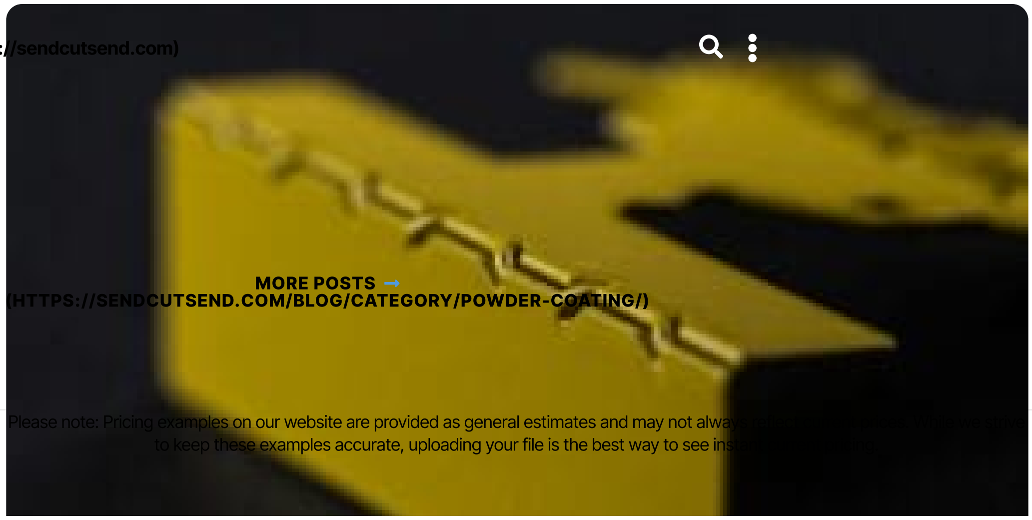

MORE POSTS → (HTTPS://SENDCUTSEND.COM/BLOG/CATEGORY/POWDER-COATING/)

Please note: Pricing examples on our website are provided as general estimates and may not alway... to keep these examples accurate, uploading your file is the best way to see ins...

(https://sendcutsend.com/blog/wave-bending-sheet-metal/)

Wave Bending Sheet Metal: Tips, Tricks, and Design Guidance (https://sendcutsend.com/blog/wave-bending-sheet-metal/)

READ MORE → (HTTPS://SENDCUTSEND.COM/BLOG/WAVE-BENDING-SHEET-METAL/)

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(https://www.instagram.com/sendcutsend) (https://www.youtube.com/channel/UCQ0t_nKCvuqsHfnWYYd1HbNw) (https://www.facebook.com/sendcutsend/) (https://www.tiktok.com/@sendcutsend)

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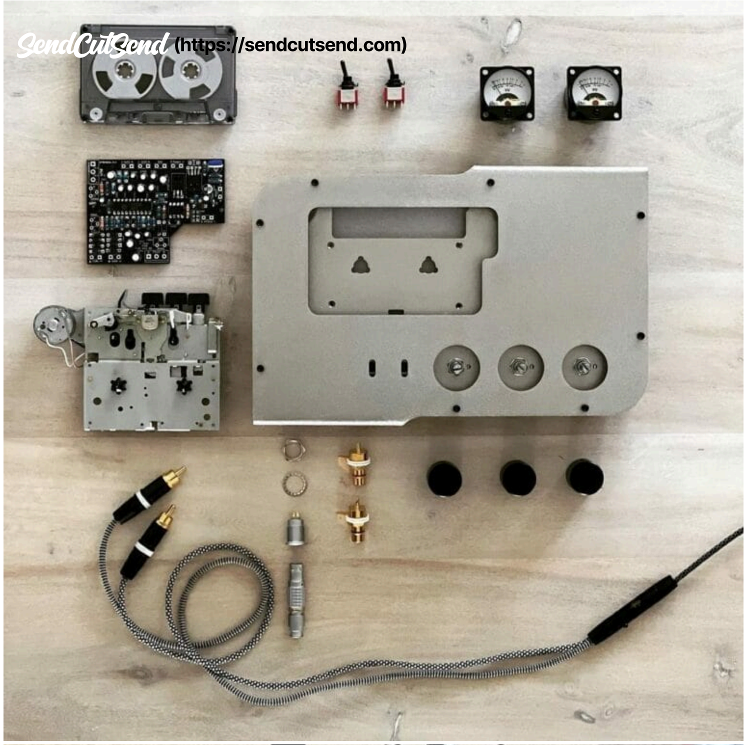


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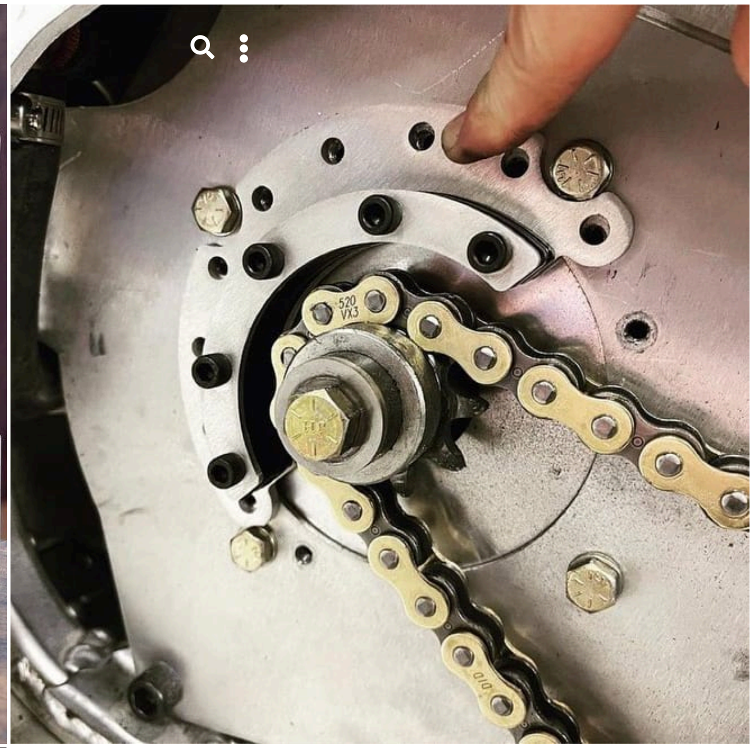

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